

MATH9102 Fundamentals of Data Analysis
Continuous Assessment
Due Date: Monday 5th January 2026 @ 23:59 (via Brightspace)
Contributes 60% to overall module marks

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Tasks

Task 1

- State a hypothesis that can be tested using **multiple linear regression**.
- It must include **at least three predictors** and incorporate a **differential effect** (i.e., an interaction or comparison effect).
- Write appropriate **R code** to build, analyse, and report on a linear regression model to test this hypothesis.
- Draw a clear **conclusion** about the hypothesis based on your findings.

Task 2

- State a hypothesis that can be tested using **binary logistic regression** with **at least three predictors**.
- Write appropriate **R code** to build, analyse, and report on a binary logistic regression model to test this hypothesis.
- Draw a clear **conclusion** based on your findings.

Task 3

- State an **additional hypothesis** that amends the hypothesis from either Task 1 or Task 2.
 - You may **add or remove a predictor** variable.
- Write appropriate **R code** to build, analyse, and report on the updated model.
- Write appropriate **R code** to **critically compare** this model with your original model (from Task 1 or Task 2, as relevant) and **report on the comparison**.
- Draw a clear **conclusion** about the revised hypothesis based on your findings.

Task 4

- Refer to the **Week 10 class on Dimension Reduction**.
- You have been provided with code that performs a dimension reduction analysis based on: *Faisal, C. M. N., Gonzalez-Rodriguez, M., Fernandez-Lanvin, D., & Andres-Suarez, J. De. (2017). Web Design Attributes in Building User Trust, Satisfaction, and Loyalty for a High Uncertainty Avoidance Culture. IEEE Transactions on Human-Machine Systems, 47(6), 847–859. <https://doi.org/10.1109/THMS.2016.2620901>*
- This dimension reduction focuses on survey items related to **interactivity, navigation, typography, and colour**.
- Complete the Quarto file provided by **analysing and reporting on the dimension reduction**.
- Draw conclusion from your findings.

Code requirements

- All work must be **reproducible in Quarto (.qmd)**.
- Code should be **clearly commented and labelled**.
- Show **all data preparation, exploration, and analysis steps**.

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What do I need to Submit?¹

Submit a **single compressed file (.zip or .rar)** named with your **student number**, containing:

- **DATA:**
 - The dataset(s) used in tasks 1, 2 and 3.
 - Either the actual dataset or a link to where your dataset can be downloaded.
 - If providing a link this can be included in your report(s).
- **REPORT:**
 - You must submit **three (3) reports**, each rendered to **.html**:
 - **Linear Regression Report** – Filename example: D1234567MLR.html
 - Addresses Task 1, and Task 3 if the extended model is linear.
 - **Logistic Regression Report** – Filename example: D1234567LOGR.html
 - Addresses Task 2, and Task 3 if the extended model is logistic.
 - **Dimension Reduction Report** – Filename example: D1234567DIMR.html
 - Addresses Task 4.

Templates are available in the Continuous Assessment section in Brightspace. You may adapt these or use your own structure.

- **CODE:**
 - You must submit **three (3) source code Quarto files**:
 - **Linear Regression** – Filename example: D1234567MLR.qmd
 - **Logistic Regression** – Filename example: D1234567LOGR.qmd
 - **Dimension Reduction** – Filename example: D1234567DIMR.qmd
- **Each file should:**
 - Contain all code used to prepare data, perform analysis, and generate results/visualizations.
 - Be sufficient to fully reproduce the corresponding report.

How do I submit?²

- You must submit using the assignment named **CA Phase TWO** in Brightspace.
 - Navigate to Assessment → Assignments

How will I be marked?³

- A detailed analytic rubric has been published with this specification and is available in Brightspace.
- A holistic version of the rubric will be attached to the assignment submission box and will be used to provide you with feedback and an indication of your level of performance using the categorisation on the rubric.

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Important Notes

- You must adhere to the conventions used in examples provided in this module. Deviations will result in deduction of marks.
- Your report **must** be self-contained. If it is not, marks will be deducted.
- You **cannot resubmit** your work once it has been marked – no second submissions will be accepted.
- **Advice and guidance** on the assignment will be provided in lectures, labs.
- Submissions sent by email for review, correction, or pre-marking in advanced of the deadline will not be accepted.
 - You may email specific questions, but not requests for pre-marking.
- Make sure your work is complete, correct, and ready before you upload it to Brightspace.

¹ You **cannot** submit a Python script. If you do it will be treated as a non-submission, and you will receive a mark of 0.

² Only submissions made using Brightspace will be marked. Email submissions will be ignored.

³ You will not receive actual marks until the module results have been reviewed by the external examiners and processed at the examination board in January 2026. You will receive clear indicators of your level of achievement.